

Firstki Aditya Fernanda

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Graduate Information Systems student at Telkom University with hands-on experience in penetration testing, vulnerability assessment, web security, networking, and secure application development. Experience in black-box security testing, plugin enumeration, vulnerability validation, threat modelling, and security analysis using the OWASP framework. Conducted thesis research across 6 virtual Moodle servers and 5 infrastructure layers, identifying and validating 122 findings including SQL Injection, XSS, IDOR, RCE, SSRF, CSRF, and container misconfigurations, and documenting them with risk ratings and remediation steps. BNSP-certified Cybersecurity Analyst, with practical experience in secure authentication, RBAC, RESTful API security, network troubleshooting, and technical mentoring.

SKILLS

- **Cybersecurity:** Penetration Testing (black-box), Vulnerability Assessment, Web Security, Network Security, Security Hardening, Threat Modeling (Attack Tree, IAAA), OWASP [Top 10], Firewall & Firewall Policy, RBAC, RESTful API Security, Database Security, Incident Analysis, Container Security.
- **Security Tools:** Nmap, Burp Suite, OWASP ZAP, Nuclei, WhatWeb, Nikto, MoodleScanner, Wireshark, [Metasploit], [Nessus], Docker, Linux
- **Standards & References:** NVD, MITRE CWE, CIS Benchmarks, [CVSS], [MITRE ATT&CK].
- **Programming:** Python, [Bash], JavaScript (ES6+), PHP, SQL, HTML5, CSS3, Node.js, Express.js, Laravel, React.js, RESTful API, WebSocket, AJAX.
- **Databases & Infrastructure:** MySQL, PostgreSQL, Redis, Git/GitHub, Docker, Postman.

CERTIFICATION

- [2026] [Certificate] BNSP Cybersecurity Analyst
- [2026] [Certificate] HackerRank. Problem Solving (Basic & Intermediate).
- [2023] [Certificate] Cyber Academy Indonesia. Web Penetration Testing
- [2022] [Certificate] DQLab. Introduction to Data with Python.

WORK EXPERIENCE

Software Engineer, *Internship.*

Jan 2025 – Jun 2025

PT Rasa Aksata Nusantara (Duluin Fintech), Bandung, West Java, Indonesia.

- Repaired 10+ critical production bugs using Node.js, maintaining system stability during high-traffic periods.
- Reviewed and resolved performance bottlenecks, and resolved reducing page load time by 25%.
- Implemented AJAX-based asynchronous data fetching, removing 4+ full page reloads per session and reducing server requests by 20%.
- Implemented RESTful API security controls across 5+ endpoints, meeting internal data encryption standards on all of them.

Mentor Cybersecurity, *Contract.*

Dec 2024 – Feb 2025

Jagoan Hosting and Practiclass, Remote, Indonesia.

- Mentored 15+ participants across 5 sessions on cybersecurity fundamentals, web security, vulnerabilities, and system hardening through hands-on exercises and practical demonstrations.
- Guided mentees through common vulnerability classes and defensive techniques, providing technical support and troubleshooting throughout the programme.
- Nominated "Most Favorite Instructor" out of 5 mentors for impact on participants' learning experience.

Cybersecurity Webinar Speaker, *Contract.*

Nov 2024 – Dec 2024

NGALUP Collaborative Network, Remote, Indonesia.

- Delivered 2+ cybersecurity webinar sessions to 20+ participants, covering Firewall & Firewall Policy, Deny-by-Default, and defence against DDoS and Botnet attacks.
- Shared preparation strategies for LKS (Lomba Kompetensi Siswa) cybersecurity and infrastructure competitions, drawing on a Silver Medal at LKS East Java 2022.
- Educated participants on fundamental cybersecurity practices and personal protection against common cyber threats.

PROJECT

Undergraduate Thesis

Jan 2026 – Jul 2026

Analysis of Plugin Vulnerabilities in Moodle-Based E-Learning Websites via Enumeration Using the OWASP Framework

- Conducted black-box penetration testing and plugin enumeration across 6 Moodle servers and 5 infrastructure layers using Nmap, OWASP ZAP, Nuclei, WhatWeb, Nikto, MoodleScanner, and other security tools.
- Identified SQL Injection, XSS, IDOR, RCE, SSRF, CSRF, security misconfigurations, and container vulnerabilities, including 48 public vulnerabilities on Moodle.
- Manually validated 122 security findings with 47 True Positives, 23 True Negatives, 0 False Positives, and 52 False Negatives, and correlated findings with NVD, MITRE CWE, and CIS Benchmarks.
- Applied OWASP, Attack Tree threat modeling, and IAAA analysis to map attack paths and develop remediation recommendations covering security hardening, access control, security headers, least privilege, and container security.

Software Engineer, Contract (part-time).

Jan 2025 – Jun 2025

Logistic Company, Fleet Management System (TRACTION).

- Designed and implemented a RESTful API with Redis caching, reducing database load by 40% and maintaining a response time of under 100ms for high-frequency location pings.
- Developed a Role-Based Access Control (RBAC) system, ensuring secure data isolation for different user levels (Driver, Manager, Admin) and preventing unauthorized access to sensitive fleet logs.
- Developed a real-time tracking module using WebSocket, enabling live updates for 50+ fleet vehicles simultaneously.
- Optimized database queries for asset tracking, resulting in 30% faster retrieval of historical trip data.

EDUCATION

Telkom University, Bandung, West Java

Sep 2022 – Sep 2026

Bachelor's degree in Information Systems | GPA 3.46/4.00

- **Relevant coursework:** Web Programming, Data Mining, and Computer Networks.
- **Other coursework project:** Inventory Management System. Landing Page, Bidding Website, and Human Resource Management System.

ORGANIZATION

Practicum Assistant

Sep 2024 – Oct 2025

Telkom University, Bandung, West Java, Indonesia.

- Led hands-on networking laboratory sessions for 3 classes (7-9 students per class).
- Designed and developed introductory Computer Network modules covering TCP/IP fundamentals, subnetting, and cabling (crimping).
- Administered practical assessments to evaluate students' understanding of network configuration and troubleshooting.
- Mentored students in network troubleshooting and foundational security awareness within infrastructure environments

Competition Division Staff

Jul 2024 – Jul 2025

Enterprise Infrastructure Management Laboratory, Bandung, West Java, Indonesia.

- Prepared 5+ lab members for 5+ cybersecurity and infrastructure competitions over one year, coordinating with 2+ cross-functional divisions.
- Researched and tested emerging security and infrastructure technologies, including container security, firewall policy, and network segmentation.
- Mentored 3+ students on web security, secure configuration, and system hardening, 2 of whom advanced to national.
- Documented 2+ internal guides on secure system design and infrastructure resilience for lab members.

Team Lead, Programming Basic Laboratory

Aug 2023 – Jan 2024

Telkom University, Bandung, West Java, Indonesia.

- Allocated task responsibilities to practicum assistants and organized a team of 6-24 assistants.
- Lectured practical work for 4 classes and guiding 7 to 9 students per class.
- Delivered report regularly to the department head or faculty supervisor.
- Responded students, practicum assistants, and faculty, ensuring queries were answered within 48 hours.

ACHIEVEMENT

- [2024] 1st Place Winner, AI App Development - International Young Entrepreneur Society Competition.
- [2024] 2nd Place Winner, CR-Youth IT by Search Telkom University.
- [2023] Best No Code Hackathon by Garuda Hacks out of 100+ teams.
- [2023] Best Performance Hacking Program Innovation by Agility Transformation out of 10+ teams.
- [2022] 2nd Place, LKS Cyber Security Competition - East Java Province.
- [2022] 4th Place, LKS Cyber Security Competition - WILKER 3, East Java Province.

LANGUAGE

- Bahasa Indonesia (Native)
- English (Intermediate)